

Hardware qualification plan: head-roll axis, frozen baseline

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Status: Proposed, not performed. Limits below are provisional project values, to be confirmed against the real application and fixture before testing.

Instrumentation (one timebase, ≥ 1 kHz): roll and yaw encoders; current target, pre-clamp command, measured current and active limit; bus voltage and PWM duty; winding/drive temperature; CAN send/receive timestamps; drive mode and fault state; original and governed references. Fixture: calibrated torque sensor, host-independent hard-wired e-stop, verified roll clearance with end stops or brake, load containment.

Global stops (any one $\rightarrow$ safe hold, end test): stale or lost feedback timestamps; wrong torque sign; unverified current/voltage convention; tracking error $> 5^\circ$; $\geq 95\%$ current utilization for > 100 ms; a latched fault the controller does not explain; temperature at the fixture limit (set below the simulated 130°C trip); motion outside verified clearance.

#	Test (in order; each gates the next)	Stage-specific stops	Acceptance to proceed
1	Identify. Axis blocked on the torque fixture: sign, $K_t (= K_\theta)$, current offset, command and feedback latency distribution, electrical convention (V_{bus} vs $V_{\text{bus}}/\sqrt{3}$).	Any measured value outside the modelled ranges: $K_t \pm 15\%$, latency > 4 ms bursts	Parameters within modelled ranges. Otherwise re-run the Task 2 margin analysis with the measured values before any motion.
2	Static holds , supported axis, derated 2.4 A: nominal payload, then known loads, at 0° and $\pm 15^\circ$.	Holding current above the governor budget $0.8 \cdot K_t I_{\text{lim}} - 0.05 \text{ N}\cdot\text{m}$	Hold error $\leq 0.5^\circ$. Loads whose calculated holding demand exceeds the budget are not attempted; the operator screens them, because the nominal-model governor cannot see the payload.
3	Slow moves , $\pm 15^\circ \rightarrow \pm 45^\circ$, nominal load, then 3.2 A.	Error $> 5^\circ$; sustained limit	$\leq 2^\circ$ RMS and $\leq 5^\circ$ peak against the governed reference; $\geq 95\%$ path progress; no faults; original-request error reported alongside.
4	Yaw disturbance , roll held: low-amplitude yaw sweep, then B/C frequencies only if the fitted coupling predicts $\geq 20\%$ current reserve.	Coupling above modelled reserve	Coupling fit validated at a held-out frequency (predicted residual within $\pm 20\%$). $\leq 2^\circ$ RMS in the admitted envelope; any yaw reduction recorded as changed delivery.
5	Faults and derating on the protected fixture: command-only, feedback-only and both-direction outages; $3.2 \rightarrow 2.4$ A derate mid-move; tracking-fault catch.	Fallback excursion beyond the verified clearance	No current target above the active limit. Re-arm only after the 50 ms fresh aligned dwell. Tracking-fault request stays suspended until replanned. Fallback excursion within clearance (simulation: up to 42° under load, so contain first).
6	Payload and envelope: the D-like shifted payload at reduced speed, then the loaded finite move.	As above	Loaded requests are reshaped or rejected as the governor reports. No request is completed by exceeding a limit.

Exit. The baseline is qualified for the tested envelope only. The adaptive correction is not tested on hardware unless it first passes a newly registered simulation comparison.